Kumari Mamta Singh

Email: mamtasinghb03@gmail.com
LinkedIn: /in/mamta-singh-b58230291/
GitHub: https://github.com/ViviBytes/
Portfolio: https://vivibytes.github.io/

Education:

Indira Gandhi Engineering College, RGPV Bhopal — *Nov 2022 – May 2026*
Bachelor of Technology-Electronics &Communications Engineering — *Sagar, Madhya Pradesh*
CGPA: 7.0

D.A.V Public School Dhudhichua — *May 2020 – April 2022*
Higher Secondary — *Singruali, Madhya Pradesh*

D.A.V Public School Khadia — *April 2019- Mar 2020*
Junior Secondary — *Saktinagaar, Uttar Pradesh*

Skills:

- *Languages:* Python (Basics), C, C++ ,SQL
- *Core Concepts:* Data Structures & Algorithms, OOP
- *Libraries/ Tools:* NumPy, Pandas, OpenCV, Matplotlib, TensorFlow (Basic), CSV, Tkinter
- *Other Tools:* Git, STL, Canvas, MS Excel
- *Domain Knowledge:* Digital Electronics, Verilog HDL (Basic)

Projects

1. Sign Language to Text/Speech System:
   - Developed a system to convert hand gestures into text and speech
   - Applied computer vision techniques for gesture recognition
   - Improved accessibility for
   - Tools Used: Python, OpenCV, NumPy
2. Hand Gesture Control System:
   - Built a touchless system to control hardware/software using hand gestures
   - Enabled real-time gesture detection and interaction
   - Enhanced human-computer interaction
   - Tools Used: Python, OpenCV, MediaPipe
3. Speech to Text/Speech System:
   - Implemented a voice-based command and response system
   - Converted speech input into text and generated audio output
   - Improved user interaction through voice control
   - Tools Used: C++, Speech Processing Libraries / APIs
4. State Guessing Game:
   - Developed an interactive map-based game for identifying states
   - Used data handling and visualization for user interaction
   - Enhanced learning through a gamed approach
   - Tools Used: Python, Pandas, Turtle.

Internship:

1. *Lanos institute*
   - C,C++ Programming
   - Practiced OOP, pointers, arrays
   - Strengthened coding and debugging skills
2. *CRISP*
   - Python Programming
   - Worked on core Python concepts
   - Built mini projects using NumPy and Pandas
3. *Aptech*
   - Data Structures & Algorithms
   - Learned arrays, linked lists, stacks, queues
   - Improved problem-solving and logic building
4. *CRISP*
   - VLSI Design
   - Learned Verilog HDL and digital circuit design.
   - Used ModelSim & Vivado for simulation.
   - Understood logic design and timing analysis

Extra-Curricular:

- Awarded on Engineers' Day for presenting an innovative start-up idea.
- Reached Semi-Finals in a Hackathon, demonstrating problem- solving and teamwork skills.
- Served as Handball Captain (School Level) and competed up to Zonal Level.
- Active member of The Next Club (College), contributing as a Video Editor.